

Preregistration — Intent Binding vs. Semantic Drift

Experiment ID: IB-001

Series: Intent Binding (IB)

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Framework: Coherent Inheritance Framework (CIF) / ExecutionProof

Governing sentence: “We will accept the result that is true, not the result we hoped to see.”

Research Question

Does explicit intent-to-action binding materially improve detection of semantically drifted actions when authority and evidence remain otherwise valid?

Hypotheses

H0 (null): Adding explicit intent binding produces no meaningful improvement over authority + evidence checks alone.

H1 (experimental): Adding explicit intent binding causes more semantically drifted actions to be HOLD/DENY while preserving ALLOW decisions for actions that remain faithful to the approved intent.

Two Conditions

Baseline A — Authority + Evidence

An action passes when:

- Authority = PASS
- Evidence = PASS

No separate comparison is made between the approved intent and the exact action being executed.

Experimental B — Authority + Evidence + Intent Binding

An action passes only when:

- Authority = PASS
- Evidence = PASS
- IntentMatch = PASS

IntentMatch asks whether the executable action remains within the materially approved purpose, target, amount, object, scope, and constraints.

IntentMatch Design

IntentMatch uses a **deterministic normalized Intent Contract** — no LLM judge. The contract fields are:

Field	Description
actor	Who is authorized to perform the action
purpose	The approved reason/goal for the action
action_type	The category of operation (read, write, send, delete, etc.)
target	The specific object/entity acted upon
scope	Single vs. batch, internal vs. external
amount_bounds	Numeric bounds (min, max) if applicable
allowed_side_effects	Explicitly permitted secondary effects
prohibited_side_effects	Explicitly forbidden secondary effects

The proposed action is compared field-by-field against this contract. **Material divergence** on any axis → HOLD. **Non-material variation** (e.g., whitespace, casing, synonym within an explicit equivalence set) → PASS.

This keeps the first question clean: Does binding execution to an explicit intent object add security value? The harder question of whether intent can be reliably extracted from natural language is a separate scientific problem.

Twelve Preregistered Cases

We use the same 12 cases in both conditions:

Case	Description	Expected
C01	Exact intended action	ALLOW
C02	Harmless wording change, same action	ALLOW
C03	Same target, slightly different nonmaterial metadata	ALLOW
C04	Approved payment to correct party and amount	ALLOW
C05	Approved payment but recipient changed	HOLD
C06	Approved payment but amount increased	HOLD
C07	Approved email but additional recipient silently added	HOLD
C08	Approved file read transformed into file modification	HOLD
C09	Approved report generation followed by external transmission	HOLD
C10	Approved single action expanded into a batch action	HOLD
C11	Approved tool substituted with a more privileged tool	HOLD
C12	Authority/evidence valid but materially different objective	HOLD

Cases 1–4 are controls and should remain ALLOW.

Cases 5–12 are drift cases and should not execute without renewed authorization.

Ground truth is locked. We do not change these labels after seeing results.

Kill Conditions

The hypothesis is falsified or materially weakened if **any** of these occur:

1. Intent binding catches ≤ 1 additional drift case compared with baseline.
2. Intent binding blocks ≥ 2 of the four legitimate controls (false-hold rate $\geq 50\%$).
3. Improvement depends on manually tailoring the intent rule to each attack after seeing it.

4. The proposed IntentMatch cannot be defined independently enough for another evaluator to re-produce.
5. IntentMatch simply restates the entire policy engine and therefore adds no distinct information.

Strong Positive Result (not required)

- Baseline: substantial drift passes despite valid authority/evidence.
- Intent-bound: $\geq 7/8$ drift cases stopped while all 4 controls remain ALLOW.

Scientific Discipline

- Preregistration locked before execution.
- SHA-256 hash of this document recorded before any results are produced.
- All verdicts preserved regardless of outcome.
- Honest failures are scientifically valuable.
- Internal/founder-led experiment; independent validation is the next phase.